

Let we have an img of dim 62×76 pixels,
and block size 16×16 .

n = pixels in y -direction m = pixels in x -direction

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dim3 dimGrid(ceil(m/16.0), ceil(n/16.0), 1);
dim3 dimBlock(16, 16, 1);
colorToGrayscaleConversion<<<dimGrid, dimBlock>>>
(Pin_d, Pout_d, m, n);

```

P_{in} = input data on global memory

P_{out} = Output on global memory

To access 2D arrays as $A[i][j]$ requires number of columns (m) to be known in compile time

This is NOT known for dynamically allocated arrays

$A_{N \times M}$ $B_{M \times K}$

$A[N][\text{changes}]$ $B[\text{changes}][K]$

$A[\text{row} * M + c]$

0
1

$B[c * K + col]$

